



VP AIR TEST NEET (2025-26)

PART TEST - 02

DROPPER

DURATION: 180 Minutes

Date: 07-09-2025

M. MARKS: 720

Topics Covered

Physics:	Motion in a Straight Line, Motion in a Plane
Chemistry:	Structure of Atom, Classification of Elements and Periodicity in Properties
Botany:	Cell Cycle and Cell Division
Zoology:	Biomolecules

GENERAL INSTRUCTIONS:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours** duration and the Test Booklet contains **180** multiple choice questions (four options with a single correct answer).
3. There is only one correct response for each question.
4. Each **correct** answer will give **4 Mark** while **1 Mark** will be **deducted** for a **wrong** answer. The maximum marks are 720.
5. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.

OMR Instructions:

1. Use only blue/black dark ballpoint pen.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

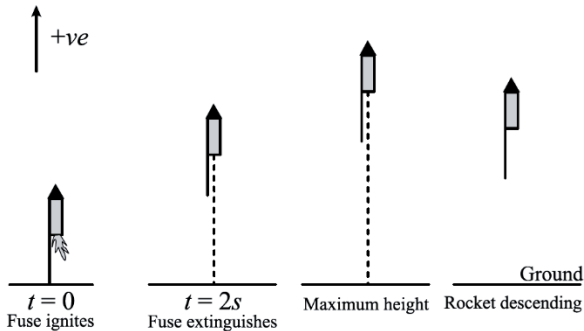
Name of the Student (In CAPITALS): _____

Roll Number: _____

OMR Bar Code Number: _____

Student's Signature: _____ **Invigilator's Signature:** _____

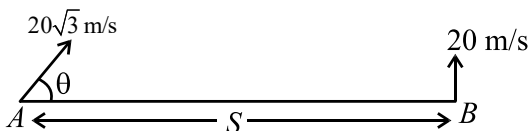
1. A Diwali rocket is launched vertically with its fuse ignited at time $t = 0$, as shown. The charge inside rocket provides constant acceleration for 2 sec. till rocket attains $V = 40 \text{ ms}^{-1}$. Afterwards rocket continues to move freely under gravity. ($g = 10 \text{ m/s}^2$)



The acceleration-time and velocity-time graphs for the rocket from launching till it reaches ground are [Take vertically upward direction as positive and assume the rocket to be a point object]

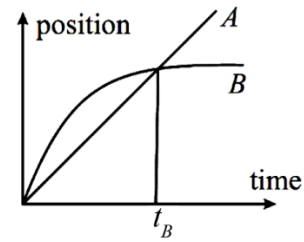
- (1)
- (2)
- (3)
- (4) None of these

2. As illustrated in the figure, two particles are projected simultaneously from points A and B, which are separated by a horizontal distance S. The particle at A is projected with an initial velocity of $20\sqrt{3} \text{ m/s}$ at an angle θ above the horizontal. The particle at B is projected vertically upwards with an initial velocity of 20 m/s . If the two particles collide after 0.5 seconds, what is the initial horizontal separation S between them? ($g = 10 \text{ m/s}^2$)

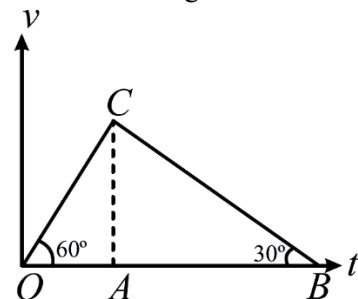


- (1) 10 m (2) $10\sqrt{2} \text{ m}$
(3) $10\sqrt{3} \text{ m}$ (4) 20 m

3. The graph shows position as a function of time for two trains running on parallel tracks. Which one of the following statements is **true**?



- (1) At time t_B , both trains have the same velocity
(2) Both trains have the same velocity at some time after t_B
(3) Both trains have the same velocity at some time before t_B
(4) Somewhere on the graph, both trains have the same acceleration
4. If $\vec{A} = \hat{i} + 2\hat{j} + 3\hat{k}$, $\vec{B} = -\hat{i} + \hat{j} + 4\hat{k}$, $\vec{C} = 3\hat{i} - 3\hat{j} - 12\hat{k}$, then find the angle between the vectors $(\vec{A} + \vec{B} + \vec{C})$ and $(\vec{A} \times \vec{B})$ in degrees.
(1) $\theta = 90^\circ$ (2) $\theta = 60^\circ$
(3) $\theta = 45^\circ$ (4) $\theta = 30^\circ$
5. Four particles move along x-axis. Their coordinates (in meters) as functions of time (in seconds) are given by
Particle 1 : $x(t) = 3.5 - 2.7 t^3$
Particle 2 : $x(t) = 3.5 + 2.7 t^3$
Particle 3 : $x(t) = 3.5 + 2.7 t^2$
Particle 4 : $x(t) = 3.5 - 3.4t - 2.7 t^2$
Which of these particles have constant acceleration?
(1) All four (2) Only 1 and 2
(3) Only 2 and 3 (4) Only 3 and 4
6. The velocity-time graph of body is shown in figure. Which of the following statements is **wrong**?



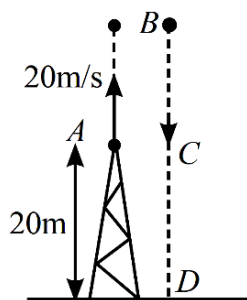
- (1) Ratio of average velocities for times OA and OB = 1
(2) $\frac{\text{time OA}}{\text{time OB}} = \frac{1}{4}$
(3) Magnitude of average accelerations for times OA and OB is in inverse of ratio of distances covered
(4) Ratio of distance covered for times OA and AB is 1 : 3

7. A man can swim with a speed of 3 m/s in still water. He wants to cross a river that is 240 m wide and flows at a speed of 5 m/s. To minimize his downstream drift, at what angle to the upstream direction must he swim?

- (1) 30° (2) 37°
(3) 53° (4) 60°

8. A particle is projected vertically upwards with speed 20 m/s from top of a tower of height 20 m as shown in figure. Given B is top most point of trajectory and C is at same height as A :

Match **List-I** with **List-II**. ($g = 10 \text{ m/s}^2$)



List-I		List-II	
A.	ratio of maximum height from ground (BD) to the initial height from ground (AD) is	I.	$\frac{1}{\sqrt{2}}$
B.	Ratio of distance travelled in 1 st second to the distance travelled in 2 nd second is	II.	1
C.	Ratio of initial speed at A to the final speed just before reaching to ground (D) is	III.	2
D.	Ratio of time taken from A to C and time taken from A to B is	IV.	3

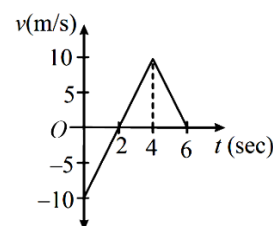
Choose the **correct** answer from the options given below:

- (1) A-III, B-IV, C-II, D-I
(2) A-III, B-IV, C-I, D-III
(3) A-I, B-IV, C-I, D-III
(4) A-I, B-III, C-I, D-II

9. Twelve persons are initially at the twelve corners of a regular polygon of twelve sides of side a . Each person now moves with a uniform speed v in such a manner that 1 is always directed towards 2, 2 towards 3, 3 towards 4, and so on. The time after which they meet is:

- (1) $\frac{v}{a}$ (2) $\frac{2a}{v}$
(3) $\frac{2a}{v(2+\sqrt{3})}$ (4) $\frac{2a}{v(2-\sqrt{3})}$

10. The figure shows the velocity-time graph for a particle moving in a straight line. If the average speed for 6 sec is ' b ' and the average acceleration from 0 sec to 4 sec is ' c ' find magnitude of bc (in m^2/s^3).

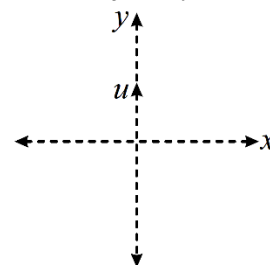


- (1) 5 (2) 20
(3) 25 (4) 40

11. Acceleration of particle moving rectilinearly is $a = 4 - 2x$ (where x is position in metre and a in m/s^2). It is at instantaneous rest at $x = 0$. At what position x will the particle again come to instantaneous rest?

- (1) 8 m (2) 4 m
(3) 2 m (4) 1 m

12. A particle is projected with speed u from O along y -axis. If acceleration of the particle is $(a\hat{i} + a\hat{j})$, then equation of trajectory of the particle is:



- (1) $y = \frac{1}{2} \frac{ax^2}{u^2}$
(2) $(y-x)^2 = \frac{2xu^2}{a}$
(3) $(x-y)^2 = \frac{2yu^2}{a}$
(4) $x^2 + y^2 = \frac{2xu^2}{a}$

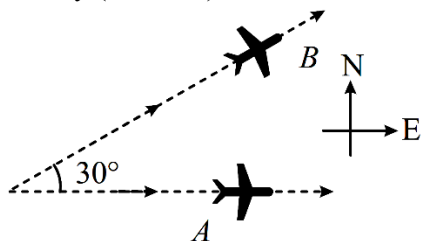
13. The velocity of a particle is $v = (v_0 + gt + Ft^2)$ m/s. Its position is $x = 0$ at $t = 0$; then its displacement (in m) after time ($t = 1$ s) is:

(1) $v_0 + g + F$ (2) $v_0 + \frac{g}{2} + \frac{F}{3}$
 (3) $v_0 + 2g + 3F$ (4) $v_0 + \frac{g}{2} + F$

14. A man is walking on a road with a velocity 3 km/hr. Suddenly rain starts falling. The velocity of rain is 10 km/hr in vertically downward direction. The magnitude of relative velocity of the rain with respect to man is:

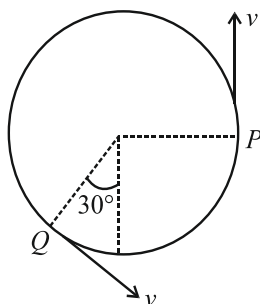
(1) $\sqrt{13}$ km/hr (2) $\sqrt{7}$ km/hr
 (3) $\sqrt{109}$ km/hr (4) 13 km/hr

15. An aeroplane A is flying horizontally due east at a speed of 400 km/hr. Passengers in A , observe another aeroplane B moving perpendicular to direction of motion of A . Aeroplane B is actually moving in a direction 30° north of east in the same horizontal plane as shown in the Figure. Determine the velocity (in km/hr) of B .



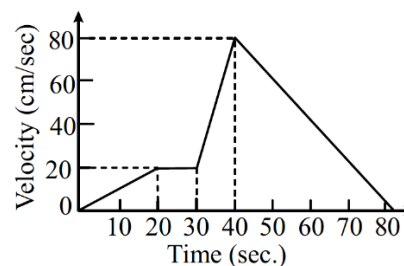
(1) $400\hat{i} + \frac{400}{\sqrt{3}}\hat{j}$ (2) $400\hat{i} + 400\hat{j}$
 (3) $400\sqrt{3}\hat{i} + 400\hat{j}$ (4) $200\sqrt{3}\hat{i} + 200\hat{j}$

16. A particle of mass 2 kg is moving in circular path with constant speed 10 m/s. The change in magnitude of velocity, when particle travels from P to Q as shown, will be [assume radius of circle is $\frac{10}{\pi^2}$]



(1) $10\sqrt{3}$ m/s (2) $20\sqrt{3}$ m/s
 (3) 10 m/s (4) 0

17. The v - t graph of a moving object is given in figure. The maximum acceleration is:



(1) 9 cm/sec^2 (2) 2 cm/sec^2
 (3) 3 cm/sec^2 (4) 6 cm/sec^2

18. From the top of a tower, a stone is thrown up. It reaches the ground in time t_1 . A second stone thrown down from the same tower with the same speed reaches the ground in time t_2 . A third stone released from rest from same tower reaches the ground in time t_3 . Then:

(1) $t_3 = \frac{t_1 + t_2}{2}$
 (2) $t_3 = \sqrt{t_1 t_2}$
 (3) $\frac{1}{t_3} = \frac{1}{t_1} - \frac{1}{t_2}$
 (4) $t_3^2 = t_1^2 - t_2^2$

19. The motion of a body is given by the equation $\frac{dv}{dt} = 6.0 - 3v$, where v is speed in m/s and t in sec.

If body was at rest at $t = 0$ choose the wrong option.

- (1) The terminal speed is 2.0 m/s
 (2) The speed varies with the time as $v(t) = 2(1 - e^{-3t})$ m/s
 (3) The speed is 0.1 m/s when the acceleration is half the initial value
 (4) The magnitude of the initial acceleration is 6.0 m/s^2

20. The vector $\hat{i} + x\hat{j} + 3\hat{k}$ is rotated through an angle θ and doubled in magnitude, then it becomes $4\hat{i} + (4x - 2)\hat{j} + 2\hat{k}$. The values of x are:

(1) $-\frac{2}{3}$
 (2) 2
 (3) $\frac{2}{3}$
 (4) both (1) and (2)

21. A particle is projected from the ground with velocity u at angle θ with horizontal. The horizontal range, maximum height and time of flight are R , H and T respectively. Now keeping u as fixed, θ is varied from 30° to 60° . Then,

- (1) R will first increase then decrease, H will increase and T will decrease
- (2) R will first increase then decrease while H and T both will increase
- (3) R will decrease while H and T will increase
- (4) R , H and T will increase

22. A body travels 102.5 m in n^{th} second and 115.0 m in $(n+2)^{\text{th}}$ second. The uniform acceleration of the body is:

- (1) 9 m/s^2
- (2) 6.25 m/s^2
- (3) 12.5 m/s^2
- (4) 5 m/s^2

23. A river is flowing from west to east at a speed of 5 meters per minute. A man on the south bank of the river, capable of swimming at 10 metres per minute in still water, wants to swim across the river in the shortest time. He should swim in a direction

- (1) due north
- (2) 30° east of north
- (3) 30° north of west
- (4) 60° east of north

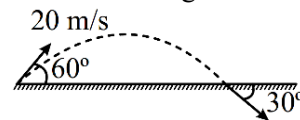
24. Two particles P and Q are located at distances r_P and r_Q respectively from the axis of a uniformly rotating disc such that $r_P > r_Q$, then:

- (1) Both P and Q have the same acceleration
- (2) Both P and Q do not have any acceleration
- (3) P has greater acceleration than Q
- (4) Q has greater acceleration than P

25. The horizontal range of a projectile is R and the maximum height attained by it is H . A strong wind now begins to blow in the direction of motion of the projectile, giving it a constant horizontal acceleration $= g/2$. Under the same conditions of projection, the horizontal range of the projectile will now be:

- (1) $R + \frac{H}{2}$
- (2) $R + H$
- (3) $R + \frac{3H}{2}$
- (4) $R + 2H$

26. A particle is projected from a horizontal surface with a velocity of 20 m/s at an angle of 60° with the horizontal. When it hits the horizontal surface, its velocity makes an angle of 30° with the horizontal. Apart from gravity, a horizontal force acts on the particle during the motion. The speed of the particle when it hits the ground is:

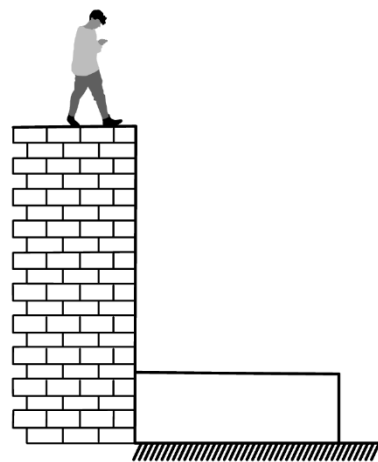


- (1) 20 m/s
- (2) $20\sqrt{3} \text{ m/s}$
- (3) 10 m/s
- (4) $10\sqrt{3} \text{ m/s}$

27. At $t = 0$ a particle leaves the origin with a velocity of 6 m/s in the positive y direction. Its acceleration is given by $\vec{a} = 2\hat{i} - 3\hat{j} \text{ m/s}^2$. The x and y coordinates of the particle at the instant the particle reaches maximum y coordinate are:

- (1) 2 m, 3 m
- (2) 4 m, 6 m
- (3) 1 m, 3 m
- (4) 2 m, 6 m

28. Suppose that a man jumps off a building 202 m high onto cushions having a total thickness of 2 m. If the cushions are crushed to a thickness of 0.5 m, what is the man's acceleration (assumed constant) when he slows down on the cushions? ($g = 10 \text{ m/s}^2$)

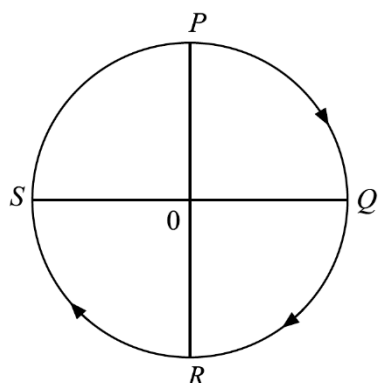


- (1) 10 m/s^2
- (2) $\frac{4000}{3} \text{ m/s}^2$
- (3) 50 m/s^2
- (4) 20 m/s^2

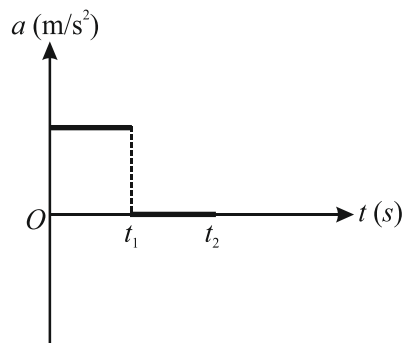
29. A helicopter rises from rest on the ground vertically upwards with a constant acceleration g . A food packet is dropped from the helicopter when it is at a height h . The time taken by the packet to reach the ground is close to [g is the acceleration due to gravity]

- (1) $t = 3.4\sqrt{\frac{h}{g}}$
- (2) $t = 1.8\sqrt{\frac{h}{g}}$
- (3) $t = \sqrt{\frac{2h}{3g}}$
- (4) $t = \frac{2}{3}\sqrt{\frac{h}{g}}$

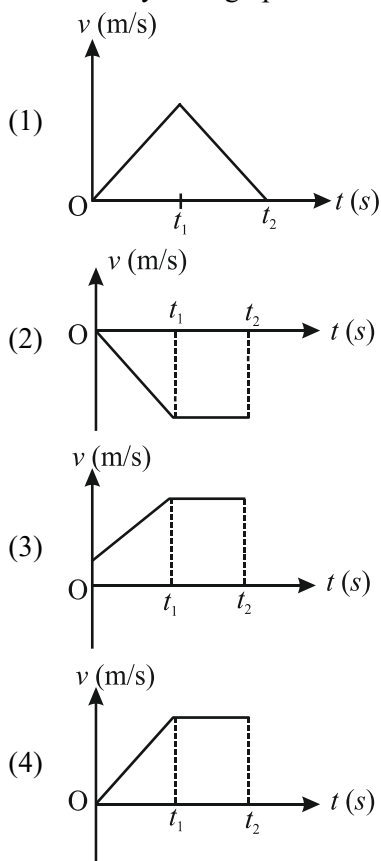
30. A cyclist starts from the point P of a circular ground of radius 2 km and travels along its circumference to the point S as shown. The displacement of cyclist is:



- (1) 6 km (2) $\sqrt{8}$ km
(3) 4 km (4) 8 km
31. The acceleration (a)-time (t) graph of a particle starting from rest and moving in straight line is shown below;



The velocity-time graph of this motion is;



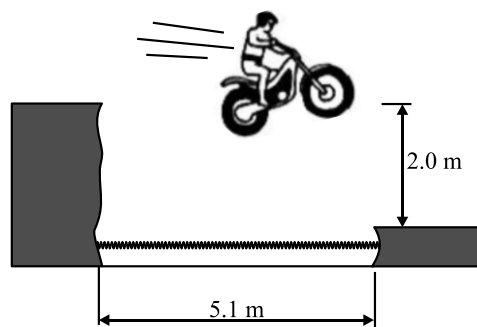
32. On a two-lane road, bike A is travelling at a constant speed of 54 km/h. Bike B is travelling behind bike A in the same lane and direction, also at 54 km/h. Simultaneously, bike C is in the adjacent lane, travelling head-on towards bike A at a speed of 72 km/h. At a certain instant, the distance AB is 200 m and the distance AC is 1 km. Bike B decides to accelerate to overtake bike A before meeting the oncoming bike C . What is the minimum acceleration of bike B required to achieve this (to avoid accident)?

- (1) 0.49 m/s^2 (2) 0.82 m/s^2
(3) 1.00 m/s^2 (4) 1.96 m/s^2

33. A body is slipping from a smooth inclined plane of height h and length l . If the angle of inclination is θ , the time taken by the body to come from the top to the bottom of this inclined plane is:

- (1) $\sqrt{\frac{2h}{g}}$ (2) $\sqrt{\frac{2l}{g}}$
(3) $\frac{1}{\sin \theta} \sqrt{\frac{2h}{g}}$ (4) $\sin \theta \sqrt{\frac{2h}{g}}$

34. A stunt motorcyclist attempts to jump horizontally across a river which is 5.1 m wide. The bank from which he will take-off is 2 m higher than the bank on which he will land, as shown below. ($g = 10 \text{ m/s}^2$)



Calculate the minimum horizontal speed he must achieve before take-off to avoid landing in the river.

- (1) 8.1 m/s
(2) 3.24 m s^{-1}
(3) 12 m s^{-1}
(4) 5.1 m/s

35. The sum of the magnitudes of two forces acting at a point is 18 and the magnitude of their resultant is 12. If the resultant is at 90° with the force of smaller magnitude, what are the, magnitudes of forces:

- (1) 12, 5 (2) 14, 4
(3) 5, 13 (4) 10, 8

36. A man at rest on a horizontal road observes that rain is falling at an angle of 45° with respect to the vertical, having a horizontal component due to the wind. He immediately starts running from rest with a constant horizontal acceleration of 1.5 m/s^2 in the same direction as the wind. After 4 seconds, he observes that the rain appears to be falling perfectly vertically. At what time, measured from the moment he started running, will the rain again appear to fall at 45° to the vertical?

- (1) 6 s (2) 8 s
(3) 10 s (4) 12 s

37. What is the displacement of a point on the rim of a wheel in contact with ground, as it completes three fourths of a revolution while rolling on the ground? (radius of wheel = R)

- (1) $2R$ (2) $\frac{R}{2}\sqrt{4+(3\pi+2)^2}$
(3) $R\sqrt{\pi^2+4}$ (4) $\frac{R}{2}\sqrt{4+(3\pi)^2}$

38. An object is moving on a circular path of radius r with constant speed v . The average acceleration of the object, after it has travelled three and half rounds, is;

- (1) $\frac{2v^2}{\pi r}$ (2) $\frac{4v^2}{3\pi r}$
(3) $\frac{4v^2}{7\pi r}$ (4) $\frac{2v^2}{7\pi r}$

39. Position of a moving particle is given by $x = 4(1 - e^{-2t}) \text{ m}$ then the initial velocity of the particle is;

- (1) 8 m/s (2) 4 m/s
(3) 10 m/s (4) Zero

40. The relation between time t and distance x for a moving body is given as $t = mx^2 + nx$, where m and n are constants. The retardation of the motion is: (Where v stands for velocity)

- (1) $2n^2v^2$ (2) $2mnv^3$
(3) $2mv^3$ (4) $2nv^3$

41. Two projectiles are thrown horizontally from a high cliff in opposite direction with same speed 20 m/s simultaneously. At what time velocities of both particles become perpendicular? ($g = 10 \text{ m/s}^2$)

- (1) $2\sqrt{2} \text{ s}$ (2) 2 s
(3) 4 s (4) $4\sqrt{2} \text{ s}$

42. Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R:
Assertion (A): To cross the river in minimum time, a swimmer should swim in a direction perpendicular to the river current.

Reason (R): When swimmer swim perpendicular to river current, then river flow helps to cross the river.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is NOT the correct explanation of A.

43. If $|\vec{A}| = 2$ and $|\vec{B}| = 4$, then match the relations in List-I with the angle θ between $\vec{A}$ and $\vec{B}$ in List-II.

List-I		List-II	
A.	$\vec{A} \cdot \vec{B} = 0$	I.	$\theta = 0^\circ$
B.	$\vec{A} \cdot \vec{B} = 8$	II.	$\theta = 90^\circ$
C.	$\vec{A} \cdot \vec{B} = 4$	III.	$\theta = 180^\circ$
D.	$\vec{A} \cdot \vec{B} = -8$	IV.	$\theta = 60^\circ$

- (1) A-III, B-II, C-I, D-IV
(2) A-II, B-I, C-IV, D-III
(3) A-II, B-IV, C-III, D-I
(4) A-III, B-I, C-II, D-IV

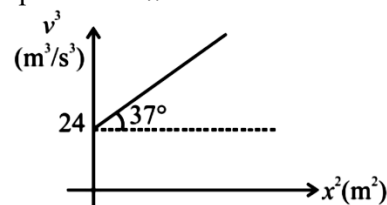
44. Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R:
Assertion (A): The position-time graph of a uniform motion in one dimension of a body can have negative slope.

Reason (R): When the speed of body decreases with time, the position-time graph of the moving body has negative slope.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is NOT the correct explanation of A.

45. For a particle moving along a straight line, graph between v^3 and x^2 is shown. The acceleration a of the particle at $x = 2 \text{ m}$ is:



- (1) $a = \frac{2}{9} \text{ m s}^{-2}$ (2) $a = \frac{1}{9} \text{ m s}^{-2}$
(3) $a = \frac{2}{3} \text{ m s}^{-2}$ (4) $a = \frac{1}{3} \text{ m s}^{-2}$

46. Number of spectrum lines obtain when electron in H-atom sample de-excite from 3rd excited state to ground state is:

(1) 3
(2) 10
(3) 8
(4) 6

47. Match List-I with List-II.

List-I (Oxide)		List-II (Example)	
(A)	Neutral oxide	(I)	P ₂ O ₅
(B)	Acidic oxide	(II)	BeO
(C)	Basic oxide	(III)	Na ₂ O
(D)	Amphoteric oxide	(IV)	NO

Choose the **correct** answer from the options given below:

(1) A-IV; B-I; C-III; D-II
(2) A-II; B-I; C-IV; D-III
(3) A-III; B-I; C-II; D-IV
(4) A-IV; B-II; C-I; D-III

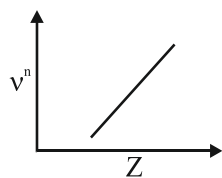
48. Given below are two statements: one is labelled as Assertion A and other is labelled as Reason R:
Assertion A: Of all the elements, hydrogen atom has the simplest line spectrum.

Reason R: Hydrogen atom has only one electron and one proton.

In the light of above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true and R is NOT the correct explanation of A.

49. Consider the following plot for characteristic X-ray spectra of elements as per Moseley, (where Z = atomic number and ν = frequency)



The value of n is:

(1) 1.5
(2) 1
(3) 0.5
(4) 2

50. Given below are two statements:

Statement I: The energy of the electron in a hydrogen atom is lower than the energy of a free electron at rest.

Statement II: Angular momentum of electron in H-atom is quantised.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

(1) Both Statement I and Statement II are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Both Statement I and Statement II are incorrect.

51. How many elements among the following are metalloids?

Ge, As, Sb, Te

(1) 1
(2) 2
(3) 3
(4) 4

52. The magnitude of velocity of electron in one electron species with increase of positive charge on the nucleus and principal quantum number:

(1) increases and decreases respectively
(2) decreases and increases respectively
(3) increases
(4) decreases

53. Among 16th group elements, minimum negative electron gain enthalpy is of:

(1) O (2) S
(3) Se (4) Te

54. A golf ball has a mass of 20 g and a speed 33.1 m s⁻¹. If the speed can be measured within accuracy of 1%, the uncertainty in the position is:

(Given: $h = 6.62 \times 10^{-34}$ Js)

(1) 5.42×10^{-34} m
(2) 7.96×10^{-33} m
(3) 1.16×10^{-35} m
(4) 1.46×10^{-32} m

55. Position of Eu ($Z = 63$) in the long form of periodic table is:

(1) 6th period and 5th group
(2) 7th period and 6th group
(3) 7th period and 4th group
(4) 6th period and 3rd group.

56. The de-Broglie's wavelength of an electron in the 3rd orbit of H-atom is:

(Given: a_0 = Bohr's radius)

- (1) $\frac{1}{3}\pi a_0$ (2) $3\pi a_0$
(3) $\frac{18}{3}\pi a_0$ (4) $9\pi a_0$

57. Metals having very low melting point are:

- (1) Ge and Li (2) In and Ba
(3) Ga and Cs (4) Si and Be

58. Consider the following statements regarding photoelectric effect:

- (A) The number of electrons ejected is proportional to the intensity of light
(B) The kinetic energy of emitted electrons increases with increase of frequency of the light used
(C) There is no time lag between the striking of light beam and the ejection of electrons from the metal surface
(D) For each metal, there is a characteristic minimum frequency, below which photoelectric effect is not observed.
(E) This effect can be explained by wave nature of electromagnetic theory.

The **correct** statements are:

- (1) A, B and D only
(2) A, B, C and D only
(3) B, C and E only
(4) A, B, C, D and E

59. IUPAC name of the element having atomic number $Z = 105$ is:

- (1) Unnilquadium
(2) Uuunpentium
(3) Ununseptium
(4) Unnilpentium

60. The **correct** increasing order of frequency for electromagnetic radiations is:

- (1) X-rays < IR-rays < long radio waves
(2) UV-rays < Microwaves < X-rays
(3) Radiowaves < UV-rays < γ -rays
(4) IR rays < Radio waves < UV-rays

61. Behaviour of beryllium is more similar with:

- (1) Sodium
(2) Boron
(3) Aluminium
(4) Silicon

62. Given below are two statements: one is labelled as Assertion A and other is labelled as Reason R:
Assertion A: Isotopes of hydrogen show same chemical behaviour.

Reason R: Isotopes of hydrogen have same number of electrons and protons.

In the light of above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true and R is NOT the correct explanation of A.

63. If the ionic radius of Mg^{2+} ion is $0.72 \text{ \AA}$, then ionic radius (in $\text{\AA}$) of Na^+ and Al^{3+} respectively are:

- (1) 1.05 and 0.99 (2) 0.66 and 0.54
(3) 0.72 and 0.72 (4) 1.02 and 0.54.

64. **Incorrect** statement regarding cathode rays is:

- (1) In the absence of electrical or magnetic field, these rays travel in the straight lines.
(2) Their behaviour cannot be observed in presence of phosphorescent material
(3) Their characteristics do not depend upon the nature of gas present in the cathode ray tube.
(4) These rays start from cathode and move towards anode.

65. Match List-I with List-II.

List-I (Electronic configuration of element)		List-II (Ionisation enthalpy in kJ mol^{-1})	
(A)	$1s^2 2s^2$	(I)	1314
(B)	$1s^2 2s^2 2p^4$	(II)	1681
(C)	$1s^2 2s^2 2p^3$	(III)	1402
(D)	$1s^2 2s^2 2p^5$	(IV)	899

Choose the *most appropriate* answer from the options given below:

- (1) A-IV; B-I; C-III; D-II
(2) A-II; B-I; C-IV; D-III
(3) A-III; B-I; C-II; D-IV
(4) A-IV; B-II; C-I; D-III

66. A certain particle carries $2.5 \times 10^{-16} \text{ C}$ of static charge. The number of electrons present in the sample is:

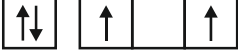
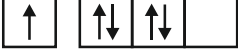
- (1) 1243 (2) 1341
(3) 1560 (4) 1728

67. On Pauling scale, electronegativity of N is similar to:
 (1) S (2) Cl
 (3) Si (4) I
68. An element with mass number 81 contains 31.7% more neutron as compared to protons. The atomic number of the element is:
 (1) 35 (2) 36
 (3) 37 (4) 34
69. Given below are two statements:
Statement I: s-block elements are found in pure state in nature.
Statement II: d-block elements mostly form colourless ions.
 In the light of the above statements, choose the *most appropriate* answer from the options given below:
 (1) Both Statement I and Statement II are correct.
 (2) Statement I is correct but Statement II is incorrect.
 (3) Statement I is incorrect but Statement II is correct.
 (4) Both Statement I and Statement II are incorrect.
70. The quantum numbers of four electrons are given below. The increasing order of their energies is:
 (A) $n = 4, \ell = 2, m_\ell = -2, m_s = -\frac{1}{2}$
 (B) $n = 3, \ell = 2, m_\ell = 1, m_s = +\frac{1}{2}$
 (C) $n = 4, \ell = 1, m_\ell = 0, m_s = +\frac{1}{2}$
 (D) $n = 3, \ell = 2, m_\ell = -2, m_s = -\frac{1}{2}$
 Choose the **correct** option:
 (1) $B = D < C < A$ (2) $B = C = D < A$
 (3) $B < D = C < A$ (4) $A < B = D < C$
71. The valence of group-17 elements is/are:
 (1) 1 only (2) 7 only
 (3) 1 and 7 (4) 8 only

72. Match List-I with List-II.

List-I (Physical quantity for one electron species)		List-II (Relation with Z)	
(A)	Radius of orbit	(I)	$\propto Z^2$
(B)	Velocity of electron	(II)	$\propto Z^{-1}$
(C)	Kinetic energy of electron	(III)	$\propto Z^{-2}$
(D)	Time period of revolution of electron	(IV)	$\propto Z$

Choose the **correct** answer from the options given below:

- (1) A-IV; B-I; C-III; D-II
 (2) A-II; B-I; C-IV; D-III
 (3) A-II; B-IV; C-I; D-III
 (4) A-IV; B-II; C-I; D-III
73. The formula of the binary compound formed by element (X) having atomic number 119 and chlorine is:
 (1) XCl (2) XCl₂
 (3) XCl₃ (4) XCl₄
74. Among the given species, Bohr's theory is applicable for how many species?
 $H^+, He, Li^+, Be^{3+}, He^+, Be^{2+}, H$
 (1) 4 (2) 5
 (3) 6 (4) 3
75. Which among the following is a chalcogen?
 (1) Fl (Z = 114) (2) Mc (Z = 115)
 (3) Lv (Z = 116) (4) Ts (Z = 117)
76. The orbital diagram in which both Hund's rule and Aufbau rule are violated is:
 (1) 
 (2) 
 (3) 
 (4) 
77. As per modern periodic law, the physical and chemical properties of the elements are periodic functions of their:
 (1) atomic masses
 (2) number of neutrons
 (3) atomic volumes
 (4) atomic numbers
78. The wave number of spectral line in the emission spectrum of hydrogen atom will be equal to $\frac{8}{9}$ times of the Rydberg's constant when electron jumps from:
 (1) $n = 2$ to $n = 1$ (2) $n = 3$ to $n = 1$
 (3) $n = 4$ to $n = 1$ (4) $n = 5$ to $n = 1$
79. As per Mendeleev's predictions, the atomic weight of Eka-aluminium is:
 (1) 56 (2) 80
 (3) 68 (4) 46

80. If de-Broglie wavelength of a particle of mass m is 100 times of its velocity, then the wavelength of particle is:
- (1) $10 \times \sqrt{\frac{m}{h}}$ (2) $\frac{1}{10} \times \sqrt{\frac{m}{h}}$
 (3) $10 \times \sqrt{\frac{h}{m}}$ (4) $\frac{1}{10} \times \sqrt{\frac{h}{m}}$
81. Given below are two statements:
Statement I: Lothar Meyer plotted the physical properties such as melting point and boiling point against atomic volume.
Statement II: Newland's law of octaves seemed to be true only for elements up to calcium.
 In the light of the above statements, choose the *most appropriate* answer from the options given below:
 (1) Both Statement I and Statement II are correct.
 (2) Statement I is correct but Statement II is incorrect.
 (3) Statement I is incorrect but Statement II is correct.
 (4) Both Statement I and Statement II are incorrect.
82. Which of the following orbitals has maximum number of radial nodes?
 (1) 5s (2) 5p_y
 (3) 4d_{yz} (4) 5d_{x²-y²}
83. **Incorrect** statement among the following is:
 (1) energy of 3d_{xy} and 3d_{x²-y²} orbitals are same
 (2) the shape of 3d_{xy} and 4d_{x²-y²} orbitals are same
 (3) Energy of 3d_{yz} and 3d_{z²} orbitals are same
 (4) The shape of 3d_{x²-y²} and 3d_{z²} are same
84. Which of the following is **not** a valid value of orbital angular momentum for an electron?
 (1) Zero (2) $\frac{h}{\sqrt{2}\pi}$
 (3) $\sqrt{\frac{3}{2}} \frac{h}{\pi}$ (4) $\sqrt{2} \frac{h}{\pi}$
85. Maximum number of electrons present in any orbit of an atom is:
 (1) $2\ell + 1$
 (2) $4\ell + 2$
 (3) $2\ell + 4$
 (4) $\ell + 4$
86. Electrons are emitted from a metal surface with zero velocity when it is exposed to radiation of wavelength 3000 Å. The threshold frequency of the metal is:
 (1) 10^{18} s^{-1}
 (2) 10^{16} s^{-1}
 (3) 10^{15} s^{-1}
 (4) 10^{17} s^{-1}
87. Which of the following groups contain non-metal only?
 (1) Group-13
 (2) Group-15
 (3) Group-18
 (4) Group-10
88. Consider the following statements regarding Heisenberg's uncertainty principle:
 (A) It is significant only for motion of microscopic objects.
 (B) It is direct consequence of wave-particle duality
 (C) In terms of energy and time, it can be expressed as $\Delta E \times \Delta t \geq \frac{h}{4\pi}$
 The **correct** statements are:
 (1) A and B only
 (2) B and C only
 (3) A and C only
 (4) A, B and C
89. 5th and 6th period in modern periodic table contain;
 (1) 18 and 32 elements respectively
 (2) 18 and 30 elements respectively
 (3) 16 and 32 elements respectively
 (4) 16 and 30 elements respectively
90. Given below are two statements: one is labelled as Assertion A and other is labelled as Reason R:
Assertion A: The ground state electronic configuration of Cr-atom is 1s² 2s² 2p⁶ 3s² 3p⁶ 3d⁵ 4s¹.
Reason R: The ground state electronic configuration of the atom of an element always corresponds to the state of the lowest total electronic energy.
 In the light of above statements, choose the **correct** answer from the options given below:
 (1) A is true but R is false.
 (2) A is false but R is true.
 (3) Both A and R are true and R is the correct explanation of A.
 (4) Both A and R are true and R is NOT the correct explanation of A.

91. During the process of development of zygote into a large multicellular organism;
- (1) meiosis and differentiation take place.
 - (2) cycles of growth and division of zygote and subsequently its daughter cells take place.
 - (3) only growth in the zygote is responsible for it.
 - (4) only one daughter cell continue to divide after each cell division, the other one does not.

92. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): Mitosis is referred to as equational division.

Reason (R): The cell organelles are equally divided amongst the daughter cells from the parent cell.

In the light of the above statements, choose the *most appropriate* answer from the options given below.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
 - (2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
 - (3) (A) is true but (R) is false.
 - (4) (A) is false but (R) is true.
93. Consider the following statements regarding the M-phase of the cell cycle.
- A. It is the most dramatic period of the cell cycle.
 - B. It involves a major reorganisation of virtually all components of the cell.
 - C. M-phase begins with cytokinesis and ends with karyokinesis.
 - D. In human cells, it lasts for about 23 hours.
 - E. It is also called G_0 phase.

In the light of above statements choose the **correct** answer from the options given below:

- (1) A, B and D are correct
 - (2) A, C and E are incorrect
 - (3) C, D and E are incorrect
 - (4) A, D and E are correct
94. During interkinesis;
- (1) amount of DNA in a cell is doubled.
 - (2) number of chromosomes is doubled.
 - (3) DNA does not replicate.
 - (4) proteins are not synthesised.

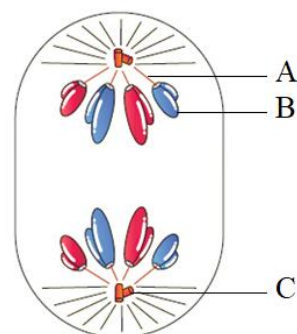
95. Match List-I with List-II.

List-I		List-II	
(A)	Zygotene	(I)	Appearance of chiasmata
(B)	Pachytene	(II)	Terminalisation of chiasmata
(C)	Diplotene	(III)	Synapsis of homologous chromosomes
(D)	Diakinesis	(IV)	Crossing over occurs

Choose the *most appropriate* answer from the options given below.

- (1) A-III, B-IV, C-I, D-II
- (2) A-III, B-I, C-IV, D-II
- (3) A-IV, B-III, C-II, D-I
- (4) A-II, B-IV, C-I, D-III

96. Go through the stage of mitosis shown in the diagram below. Select the **correct** option regarding it.



- (1) This stage occurs only after the parent chromosomes have decondensed.
- (2) 'B' is one of the daughter chromosomes.
- (3) 'C' are responsible for formation of equatorial plate in this stage.
- (4) Elongation of 'A' leads to movement of chromosome away from the equator.

97. Given below are two statements:

Statement I: The female gamete of onion cell has 8 chromosomes.

Statement II: The haploid onion root tip cell has 16 chromosomes.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Both Statement I and Statement II are correct.
- (2) Both Statement I and Statement II are incorrect.
- (3) Statement I is correct but Statement II is incorrect.
- (4) Statement I is incorrect but Statement II is correct.

98. While studying cell division in a somatic cell, the scientist observed the movement of centrioles towards opposite poles. Identify the **true** statement about it.

- (1) Bivalents have separated.
- (2) DNA replication is yet to take place.
- (3) Each chromosome consists of four chromatids.
- (4) The chromosomes will soon line up the equator.

99. Which one of the following pairs is **incorrectly** matched?

- (1) Leptotene - Compaction of chromosomes
- (2) Pachytene - Activity of recombinase enzyme
- (3) Diakinesis - Disappearance of nucleolus
- (4) Zygotene - Formation of tetrad of cells

100. Arrange the following events of meiosis II in the **correct** sequence.

- A. Formation of daughter chromosomes
- B. Formation of mitotic apparatus
- C. Division of cytoplasm
- D. Alignment of chromosomes on equator
- E. Attachment of spindle fibres to kinetochores of chromosomes

Choose the **correct** option.

- (1) B → E → D → A → C
- (2) B → D → E → A → C
- (3) B → E → D → C → A
- (4) B → A → E → D → C

101. The heart cells in adult animals that do **not** appear to exhibit division;

- (1) are present in a metabolically inactive state.
- (2) keep on synthesizing DNA continuously.
- (3) have exited G₁ phase of cell cycle.
- (4) become dormant and eventually undergo death.

102. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): In yeast, the interphase lasts about 85.5 minutes of the total duration cell cycle.

Reason (R): Yeast can progress through the cell cycle in only about 90 minutes.

In the light of the above statements, choose the *most appropriate* answer from the options given below.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
- (3) (A) is true but (R) is false.
- (4) (A) is false but (R) is true.

103. The gamete of a cell has 20 chromosomes. The number of bivalents in its parent cell would be;

- (1) 40
- (2) 10
- (3) 20
- (4) 30

104. Which of the following events occur during the Telophase stage of mitosis?

- A. Chromosomes cluster at opposite spindle poles.
- B. Chromosomes lose their identity as discrete elements.
- C. Nuclear envelope assembles around the chromosome clusters.
- D. Nucleolus, golgi complex and ER reappear.
- E. Centromeres split and chromatids separate.

Choose the **correct** option.

- (1) A, B, C only
- (2) A, B, C, D only
- (3) C, D, E only
- (4) B, C, E only

105. Match List-I with List-II.

List-I		List-II	
(A)	Interphase	(I)	5 stages
(B)	Karyokinesis	(II)	3 sub stages
(C)	Prophase I	(III)	2 major phases
(D)	Cell cycle	(IV)	4 phases

Choose the *most appropriate* answer from the options given below.

- (1) A-III, B-I, C-II, D-IV
- (2) A-II, B-IV, C-I, D-III
- (3) A-III, B-IV, C-II, D-I
- (4) A-IV, B-III, C-I, D-II

106. The formation of a cell-plate is characteristic of cytokinesis in:

- (1) animal cells, where it forms in the centre and grows outwards.
- (2) plant cells, where it represents the middle lamella.
- (3) animal cells, where it is a precursor to the cleavage furrow.
- (4) plant cells, where it grows inwards from the periphery.

107. Identify the **correct** statements regarding the prophase of mitosis.

- A. Chromosomal material becomes untangled.
- B. Chromatin undergoes condensation.
- C. By its end, nucleolus becomes visible.
- D. It is the first stage of nuclear division.
- E. It occurs in prokaryotic cells.

Choose the **correct** option.

- (1) A, B, D
- (2) A, B, C, E
- (3) C, D, E
- (4) A, C, D

108. Given below are two statements:

Statement I: A very significant contribution of mitosis is cell repair.

Statement II: Mitotic divisions in the meristematic tissues result in a continuous growth of higher plants throughout their life.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Both Statement I and Statement II are correct.
- (2) Both Statement I and Statement II are incorrect.
- (3) Statement I is correct but Statement II is incorrect.
- (4) Statement I is incorrect but Statement II is correct.

109. What will be the consequence if mitosis in all the cells of a person stops?

- (1) The gametes will not be formed.
- (2) Variations in progeny will not be seen.
- (3) The lining of gut will soon get dissolved.
- (4) The dead cells will be replaced by new ones.

110. The mode of formation of gametes in male honey bee is;

- (1) reductional division.
- (2) equational division.
- (3) fission.
- (4) meiosis I followed by meiosis II.

111. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): In oocytes of some vertebrates, the diplotene stage can last for months or years.

Reason (R): The dissolution of the synaptonemal complex and terminalisation of chiasmata is a very slow process in these oocytes.

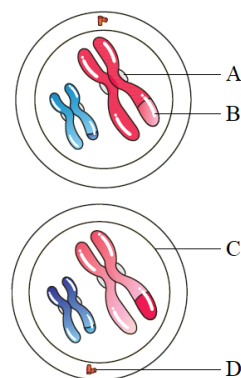
In the light of the above statements, choose the *most appropriate* answer from the options given below.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
- (3) (A) is true but (R) is false.
- (4) (A) is false but (R) is true.

112. In which phase is the meiotic spindle assembled to prepare the homologous chromosomes for separation?

- (1) Prophase I
- (2) Metaphase I
- (3) Prophase II
- (4) Metaphase II

113. Identify the **correct** characteristic of the labels shown for a stage of cell cycle below.



Choose the **correct** option.

- (1) 'A' is the site of crossing over.
- (2) 'B' is the site of attachment of microtubules from the spindle fibres.
- (3) 'C' is the envelope that surrounds the segregated homologous chromosomes.
- (4) 'D' disintegrates in this stage.

114. Select the **correctly** matched pair.

- (1) S phase - DNA replicates in cytoplasm
- (2) Diplotene - Synapsis of homologous chromosomes
- (3) Meiosis II - Reduction division
- (4) Anaphase II - Splitting of centromere

115. How many of the following events do **not** occur during interphase?

- A. DNA synthesis
- B. Formation of cell furrow
- C. Centriole duplication
- D. Synthesis of proteins
- E. Duplication of organelles

Choose the **correct** option.

- (1) Two
- (2) Three
- (3) One
- (4) Five

116. Given below are two statements:

Statement I: Interkinesis is a stage between meiosis I and meiosis II.

Statement II: During interkinesis, the chromosomes reach extreme state of dispersion.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Both Statement I and Statement II are correct.
- (2) Both Statement I and Statement II are incorrect.
- (3) Statement I is correct but Statement II is incorrect.
- (4) Statement I is incorrect but Statement II is correct.

117. The cell organelles like plastids are distributed to the daughter cells during;
- (1) karyokinesis.
 - (2) diakinesis.
 - (3) cytokinesis.
 - (4) interkinesis.

118. Choose the **incorrect** statement.

- (1) Cell growth in terms of cytoplasmic increase is a continuous process.
- (2) The replicated chromosomes are distributed to daughter nuclei by a simple series of events during cell division.
- (3) The events of cell cycle are under genetic control.
- (4) Cell division is a very important process in all living organisms.

119. Arrange the following in the descending order of their durations.

- A. Interphase
- B. Prophase I
- C. Meiosis I
- D. Prophase II

Choose the **correct** option.

- (1) $C \rightarrow A \rightarrow B \rightarrow D$
- (2) $C \rightarrow B \rightarrow A \rightarrow D$
- (3) $A \rightarrow C \rightarrow B \rightarrow D$
- (4) $A \rightarrow B \rightarrow C \rightarrow D$

120. Meiosis;

- (1) causes genetic variability but fails to conserve the chromosome number.
- (2) conserves the specific chromosome number across generations in sexually reproducing organisms.
- (3) is the primary method of growth in multicellular organisms.
- (4) restores the diploid phase in the life cycle of sexually reproducing organisms.

121. Match List-I with List-II.

List-I		List-II	
(A)	Asters	(I)	Primary constriction holding sister chromatids
(B)	Spindle fibres	(II)	Form mitotic apparatus along with asters
(C)	Centromere	(III)	Plane of alignment of chromosomes during 2 nd stage of nuclear division
(D)	Metaphase plate	(IV)	Radiate out from centrosomes

Choose the *most appropriate* answer from the options given below.

- (1) A-IV, B-II, C-I, D-III
- (2) A-II, B-IV, C-I, D-III
- (3) A-IV, B-I, C-II, D-III
- (4) A-I, B-II, C-IV, D-III

122. Given below are two statements:

Statement I: The daughter cells formed after meiosis during spermatogenesis in human females are equal in size.

Statement II: The daughter cells formed after meiosis during oogenesis in human males are unequal in size.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Both Statement I and Statement II are correct.
- (2) Both Statement I and Statement II are incorrect.
- (3) Statement I is correct but Statement II is incorrect.
- (4) Statement I is incorrect but Statement II is correct.

123. The chiasmata;

- (1) are formed during anaphase of meiosis I.
- (2) are X-shaped structures.
- (3) represent the sites of crossovers in non-homologous chromosomes.
- (4) are not formed during reduction division.

124. By the end of mitotic prophase;

- (1) nuclear membrane disintegrates.
- (2) the chromosomes are not visible distinctly.
- (3) golgi complexes and endoplasmic reticulum are visible.
- (4) daughter chromosomes move towards same pole.

125. The formation of a syncytium;

- (1) occurs when cytokinesis is followed by karyokinesis.
- (2) occurs when karyokinesis is followed by cytokinesis.
- (3) is seen in the liquid endosperm of coconut.
- (4) leads to enucleated conditions in cells.

- 126.** Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): In plant cells, the cell wall cannot be extended towards the centre during cytokinesis.

Reason (R): Plant cell membrane is a rigid and inextensible structure.

In the light of the above statements, choose the *most appropriate* answer from the options given below.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
- (3) (A) is true but (R) is false.
- (4) (A) is false but (R) is true.

- 127.** Which phase of the cell cycle represents the interval between mitosis and the initiation of DNA replication?

- (1) S phase
- (2) G₂ phase
- (3) M phase
- (4) G₁ phase

- 128.** _____ represents transition to metaphase.

- (1) Diplotene
- (2) Diakinesis
- (3) Pachytene
- (4) Leptotene

- 129.** After splitting of the centromere;

- A. Chromatids separate thus forming daughter chromatids.
- B. The daughter chromatids are also called daughter chromosomes.
- C. During migration, the centromere of each daughter chromosome is directed towards the pole.
- D. The arms of the daughter chromosomes trail behind the centromere.
- E. The daughter chromosomes are composed of two chromatids each.

Which of the following statements are **correct**?

- (1) A, B, C and D
- (2) A, B, C and E
- (3) B, C, D and E
- (4) A, C, D and E

- 130.** Identify the **correct** statement about quiescent stage of cell cycle.

- (1) All human cells are present in this stage.
- (2) The cells of this stage can never divide.
- (3) It represents the phase between two successive M phases.
- (4) It is said to be an inactive stage of cell cycle.

- 131.** Match **List-I** with **List-II**.

List-I		List-II	
(A)	Sister chromatids separate	(I)	Cytokinesis after telophase II
(B)	Four haploid cells are formed	(II)	Prophase I
(C)	Bivalents arrange on the equatorial plate	(III)	Anaphase II
(D)	Homologous chromosomes pair to form bivalents	(IV)	Metaphase I

Choose the *most appropriate* answer from the options given below.

- (1) A-III, B-I, C-IV, D-II
- (2) A-I, B-III, C-IV, D-II
- (3) A-III, B-IV, C-I, D-II
- (4) A-II, B-I, C-IV, D-III

- 132.** The gamete of a cell has 20 pg DNA. The amount of DNA (in pg) in G₂ phase, metaphase I and prophase II, respectively, of its parent cell would have been;

- (1) 40, 40, 20
- (2) 80, 40, 80
- (3) 80, 80, 40
- (4) 40, 20, 40

- 133.** In the resting phase of cell cycle during meiosis, the;

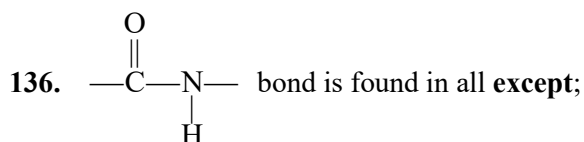
- (1) size of the cell is reduced by half.
- (2) number of cell organelles is reduced by half.
- (3) chromosome number is doubled.
- (4) DNA content is doubled.

- 134.** Haploid cells divide by mitosis in;

- (1) some lower plants.
- (2) only higher plants.
- (3) none of the animals.
- (4) majority of animals.

- 135.** What is the chromosome count at a single pole of a cell at the end of Anaphase I, when compared to the chromosome count of the original diploid parent cell?

- (1) Double
- (2) Half
- (3) Same
- (4) One fourth



- (1) Polypeptides
- (2) Proteinaceous enzymes
- (3) Fatty acid
- (4) Proteins

137. Match List-I with List-II.

List-I (Category)		List-II (Secondary Metabolites)	
(A)	Toxins	(I)	Concanavalin A
(B)	Terpenoides	(II)	Monoterpenes, Diterpenes
(C)	Alkaloids	(III)	Morphine, Codeine
(D)	Lectins	(IV)	Abrin, Ricin

Choose the **correct** answer from the options given below:

- (1) A-II, B-III, C-I, D-IV
- (2) A-II, B-IV, C-III, D-I
- (3) A-IV, B-II, C-III, D-I
- (4) A-IV, B-III, C-II, D-I

138. Given below are two statements.

Statement I: Enzymes are highly specific, with a single enzyme typically acting upon only one specific substrate.

Statement II: A single substrate molecule can be acted upon by multiple different enzymes in different reactions/pathways.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

139. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Unsaturated fats are more reactive compared with the saturated fats.

Reason R: Unsaturated fats possess no C=C double bonds, which are more reactive than single bonds.

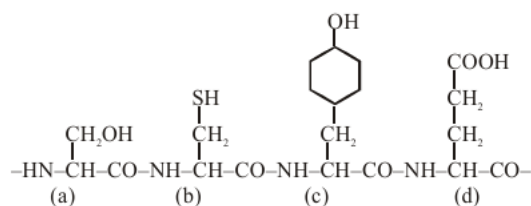
In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

140. Which of the following statements about amino acids is **incorrect**?

- (1) The human body can synthesize all 20 types of amino acids required for protein synthesis, making dietary protein intake non-essential.
- (2) Amino acids such as valine and leucine must be acquired through the diet.
- (3) The R-group of aspartic acid contains an additional carboxyl group, conferring acidic properties.
- (4) Lysine and arginine are basic amino acids.

141. Recognise the figure and find out the **correct** matching: -



- (1) (a)-cysteine, (d)-tyrosine, (c)-glutamic acid, (b)-serine
- (2) (b)-cysteine, (c)-tyrosine, (d)-glutamic acid, (a)-serine
- (3) (d)-cysteine, (a)-tyrosine, (b)-glutamic acid, (c)-serine
- (4) (c)-cysteine, (b)-tyrosine, (a)-glutamic acid, (d)-serine

142. Select the **incorrect** match.

- (1) RuBisCO $\rightarrow$ Most abundant protein in whole biosphere
- (2) Collagen $\rightarrow$ Most abundant protein in animal world
- (3) Melting of ice $\rightarrow$ Chemical change
- (4) Chemical reaction $\rightarrow$ Involves breaking and making of new bonds

143. Read the following statements (A to D) and choose the option which states them as true (T) or False (F) **correctly**.

- A. Barium hydroxide, a strong base, reacts with sulfuric acid, a strong acid, to produce barium sulphate, a salt, and water.
- B. When enzyme catalysed reactions are observed, the rate would be vastly higher than an uncatalysed reaction.
- C. A multistep chemical reaction, when each of the steps is catalysed by the same enzyme complex only, is called a metabolic pathway.
- D. Chemical stability is an important quality of thermostable enzymes isolated from thermophilic organisms.

A B C D

- (1) F F F F
- (2) T T F F
- (3) F F F T
- (4) T F T F

144. The enzyme carbonic anhydrase in red blood cells catalyzes the conversion of carbonic acid (H_2CO_3) into CO_2 and H_2O . In a laboratory experiment, it was observed that 600,000 molecules of CO_2 (product) were formed in one minute. What is the rate of this enzymatic reaction, expressed in molecules of product formed per second?
- 100 molecules/second
 - 1,000 molecules/second
 - 10,000 molecules/second
 - 600,000 molecules/second

145. Given below are two statements.

Statement I: Glucose, fatty acids and amino acids are not primary metabolites.

Statement II: Similar to amino acids, fatty acids are also classified into essential and non-essential fatty acids.

In the light of the above statements, choose the most appropriate answer from the options given below:

- Statement I is correct but Statement II is incorrect.
- Statement I is incorrect but Statement II is correct.
- Both Statement I and Statement II are correct.
- Both Statement I and Statement II are incorrect.

146. Match List-I with List-II.

List-I		List-II	
(A)	Cholesterol	(I)	Pentose sugar + Nitrogenous base
(B)	Adenylic acid	(II)	A component of animal cell membrane
(C)	Adenosine	(III)	Non-essential amino acid
(D)	Proline	(IV)	Pentose sugar + Nitrogenous base + Phosphate group

Choose the **correct** answer from the options given below:

- A-II, B-I, C-III D-IV
- A-II, B-III, C-IV, D-I
- A-I, B-II, C-III, D-IV
- A-II, B-IV, C-I, D-III

147. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Enzymes have a specific active site that binds to the reactive site on a substrate molecule.

Reason R: The interaction between the active and reactive sites causes the enzyme and substrate to move away from each other.

In the light of the above statements, choose the **correct** answer from the options given below:

- Both A and R are true and R is the correct explanation of A.
- Both A and R are true but R is NOT the correct explanation of A.
- A is true but R is false.
- A is false but R is true.

148. Consider the following statements regarding the metabolic fates of glucose through the glycolytic pathway:

- Glycolysis is a universal pathway that, under aerobic conditions in human somatic cells, concludes with the formation of pyruvic acid.
- In human skeletal muscle during intense exercise, the end product of this pathway is lactic acid.
- In yeast, the process of fermentation involves the conversion of pyruvic acid to ethanol and CO_2 , a process that does not occur in human muscle cells.
- The final end product of the glycolytic pathway is always the same, regardless of the organism or the presence of oxygen.

Which of the following combinations includes all the **correct** statements?

- A and C only
- A, B, and C only
- B and D only
- A, B, C, and D

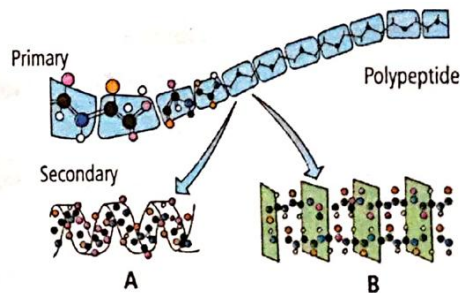
149. A biochemical reagent is designed to give a positive result by reacting specifically with a carbonyl group (aldehyde/ketone only) ($>\text{C}=\text{O}$). Which of the following combinations of biomolecules would all yield a positive test with this reagent?

- Glucose, an amino acid, and a fatty acid
- Fructose, a fatty acid, and a phospholipid
- Glucose, fructose, and a nucleotide
- Glucose, fructose, and pyruvic acid

150. When preparing a salad dressing, oil and vinegar (which is mostly water) are mixed together. Despite vigorous shaking, the oil droplets eventually separate and form a distinct layer on top of the vinegar. This phenomenon occurs because:

- The fatty acid chains of the oil are primarily hydrophilic and form strong hydrogen bonds with each other, excluding the water.
- The water molecules are nonpolar and cannot form effective interactions with the polar triglyceride molecules of the oil.
- The triglyceride molecules in the oil are largely nonpolar and cannot form hydrogen bonds with polar water molecules, leading to their exclusion and aggregation.
- The ester bonds in the oil are hydrolyzed by water, causing the oil to precipitate out of the solution.

151. Refer to the given figure.



Formation of structures A and B could be due to:

- (1) Formation of peptide bonds and linking together of two or more polypeptides respectively
- (2) Formation of hydrogen bonds and linking together of two or more polypeptides respectively
- (3) Formation of hydrogen bonds and formation of hydrogen bonds respectively
- (4) Formation of peptide bonds and formation of peptide bonds respectively

152. In protoplasm, the amount of organic compounds, in decreasing order, are:

- (1) Protein, Nucleic acid, Carbohydrate, Lipid
- (2) Water, Protein, Nucleic acid, Lipid
- (3) Carbohydrate, Protein, Lipid, Nucleic acid
- (4) Protein, Carbohydrate, Nucleic acid, Lipid

153. Given below are two statements.

Statement I: Out of 20 amino acids glycine is the only one having no asymmetric carbon.

Statement II: Some competitive inhibitors are often used in the control of bacterial pathogens.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

154. Study the given statements and select the **correct** answer.

- I. Cellulose is a homopolymer of glucose.
 - II. Inulin is a homopolymer of fructose.
 - III. Starch gives blue colour with iodine solution.
 - IV. Nucleic acids are heteropolymer of nucleotides.
- (1) Statements I, II and III are correct.
 - (2) Statements I, II and IV are correct.
 - (3) Statements II and III are correct.
 - (4) All statements are correct.

155. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Despite being a disaccharide composed of two monosaccharides (glucose and fructose) both reducing, sucrose is a non-reducing sugar.

Reason R: The glycosidic bond in sucrose is formed between the (C-1) of α -glucose and the (C-2) of β -fructose.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Both A and R are true, and R is the correct explanation of A.
- (2) Both A and R are true, but R is not the correct explanation of A.
- (3) A is true, but R is false.
- (4) A is false, but R is true.

156. Match List-I with List-II.

List-I		List-II	
(A)	Ions	(I)	A component of cell membrane
(B)	Lipids	(II)	Genetic material
(C)	Lecithin	(III)	Form 1% of the total cellular mass
(D)	DNA and RNA	(IV)	Not strictly macromolecule

Choose the **correct** answer from the options given below:

- (1) A-IV, B-III, C-II, D-I
- (2) A-III, B-II, C-IV, D-I
- (3) A-III, B-IV, C-I, D-II
- (4) A-III, B-IV, C-II, D-I

157. Read the following statements and select the option that **correctly** sorts these with respect to A and B Where A is acid-soluble pool and B is acid insoluble pool.

- I. Molecular weight ranging from 18 to 800 Daltons (Da) approximately
- II. Proteins, nucleic acids, polysaccharides and lipids
- III. Contain chemicals that have molecular weight of 10,000 Da and above
- IV. Contain monomeric substances
- V. Generally, contain polymeric substances

A	B
(1) I, II, III	IV, V
(2) II, IV	I, III, V
(3) I, IV	II, III, V
(4) I, III, V	II, IV

- 158.** When a protein is denatured by extreme heat or pH, it undergoes several structural changes that lead to a loss of its biological function. All of the following are characteristic changes of denaturation **except**:
- (1) Disruption of hydrogen bonds that stabilize α -helices and β -sheets.
 - (2) Loss of the specific three-dimensional folding of the polypeptide chain.
 - (3) Hydrolysis of the peptide bonds, leading to the breakdown of the protein into individual amino acids.
 - (4) Disruption of hydrophobic interactions and ionic bonds between distant R-groups.
- 159.** Given below are two statements.
Statement I: Purines are heterocyclic and have 5 nitrogens in their ring.
Statement II: Pyrimidines have a hexagonal ring and have 2 nitrogens in their ring.
 In the light of the above statements, choose the most appropriate answer from the options given below:
- (1) Statement I is correct but Statement II is incorrect.
 - (2) Statement I is incorrect but Statement II is correct.
 - (3) Both Statement I and Statement II are correct.
 - (4) Both Statement I and Statement II are incorrect.
- 160.** Despite their diverse chemical structures, what is the primary functional role that enzymes, vitamins, and hormones share in a living organism?
- (1) They all act as the primary source of energy for cellular activities.
 - (2) They are all stored in the body for long-term use.
 - (3) They all participate in the control and regulation of metabolic pathways.
 - (4) They all serve as structural components of cells.
- 161.** Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Malonate, which structurally resembles succinate, inhibits the enzyme succinic dehydrogenase.
Reason R: The inhibitor binds to an allosteric site, changing the conformation of the enzyme's active site.
 In the light of the above statements, choose the **correct** answer from the options given below:
- (1) A is true but R is false.
 - (2) A is false but R is true.
 - (3) Both A and R are true and R is the correct explanation of A.
 - (4) Both A and R are true but R is not the correct explanation of A.
- 162.** Which of the following pairs is **correctly** matched?
- (1) Holoenzyme $\rightarrow$ The protein part of an enzyme, catalytically inactive.
 - (2) Ribozyme $\rightarrow$ An enzyme where the catalytic activity resides in its polypeptide chain.
 - (3) Co-enzyme $\rightarrow$ An organic, non-protein molecule that is transiently bound to an enzyme and is often derived from vitamins.
 - (4) Prosthetic group $\rightarrow$ A metal ion, such as Mg^{2+} or Fe^{2+} , that is essential for enzyme function.
- 163.** Which of the following statements about enzymes is **false**?
- (1) Enzymes increase the rate of a reaction by providing an alternative pathway with a lower activation energy.
 - (2) The function of an enzyme is directly related to its specific three-dimensional structure.
 - (3) Enzymes alter the overall free energy change (ΔG) of a reaction, making it more favourable.
 - (4) End product of a reaction influences the feedback inhibition of an enzyme.
- 164.**
$$C_{12}H_{22}O_{11} + H_2O \xrightarrow{\text{Enzyme A}} 2C_6H_{12}O_6$$

Maltose
Glucose

 Enzyme A used in this reaction, belongs to which class of enzymes?
- (1) Dehydrogenases
 - (2) Transferases
 - (3) Hydrolases
 - (4) Lyases
- 165.** Which of the following statements **correctly** differentiates between natural gums and synthetic adhesives like Fevicol?
- (1) Natural gums are proteins derived from plants, while Fevicol is a polysaccharide.
 - (2) Both natural gums and Fevicol are polysaccharides, but Fevicol is a synthetic version.
 - (3) Natural gums are polysaccharides of plant origin, whereas Fevicol is a synthetic polymer like polyvinyl acetate.
 - (4) Natural gums are used for their adhesive properties, while Fevicol is primarily used as a food thickener.
- 166.** Given below are two statements.
Statement I: The organic compound which form 5-7% of the total cellular mass in living organisms, have its monomeric units joined together with the help of phosphodiester bonds.
Statement II: Calcium, magnesium, carbon dioxide and sulphur can be found in the remains of a dry living tissue burnt fully.
 In the light of the above statements, choose the most appropriate answer from the options given below:
- (1) Statement I is correct but Statement II is incorrect.
 - (2) Statement I is incorrect but Statement II is correct.
 - (3) Both Statement I and Statement II are correct.
 - (4) Both Statement I and Statement II are incorrect.

167. A biochemist wants to create a series of fats with different melting points. Which of the following **correctly** ranks the fatty acids from the one that would produce a fat with the lowest melting point to the one with the highest melting point?

- (1) Linolenic acid (18C, 3 double bonds) < Stearic acid (18C, saturated) < Palmitic acid (16C, saturated)
- (2) Stearic acid (18C, saturated) < Palmitic acid (16C, saturated) < Linolenic acid (18C, 3 double bonds)
- (3) Linolenic acid (18C, 3 double bonds) < Palmitic acid (16C, saturated) < Stearic acid (18C, saturated)
- (4) Palmitic acid (16C, saturated) < Stearic acid (18C, saturated) < Linolenic acid (18C, 3 double bonds).

168. Consider the following statements:

- A. Elemental analysis shows that living tissues contain unique elements that are entirely absent in the earth's crust.
- B. The key difference in elemental composition between living organisms and the earth's crust lies in the relative abundance of carbon and hydrogen.
- C. The chemical analysis method of a living tissue uses trichloroacetic acid to separate cellular components.
- D. Ash test is a destructive experiment.

How many of the above statements are **correct**?

- (1) One
- (2) Two
- (3) Three
- (4) Four

169. The enzyme commission number of an enzyme 'Y' is 2.7.1.1. Based on this classification, 'Y' is an enzyme that catalyses which of the following reactions?

- (1) Glucose + ATP → Glucose-6-Phosphate + ADP
- (2) A dipeptide + H₂O → Two amino acids
- (3) Pyruvate → Acetaldehyde + CO₂
- (4) S reduced + S' oxidised → S oxidised + S' reduced.

170. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Haemoglobin is an example of a protein with a quaternary structure.

Reason R: Adult haemoglobin is composed of four different types of polypeptide subunits, that assemble into a functional complex.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

171. If the formation of peptide bond involves the removal of a water molecule, what is the molecular formula of a tripeptide formed from three molecules of alanine, given that the formula for alanine is C₃H₇NO₂?

- (1) C₉H₂₁N₃O₆
- (2) C₉H₁₉N₃O₅
- (3) C₉H₁₇N₃O₄
- (4) C₉H₁₅N₃O₃

172. Match **List-I** with **List-II**.

List-I		List-II	
(A)	GLUT-4	(I)	27.7% of weight of Earth crust
(B)	Antibody	(II)	Represents inorganic constituents of living tissue
(C)	Silicon	(III)	Fights infectious agents
(D)	NaCl, CaCO ₃ , PO ₄ ³⁻ , SO ₄ ²⁻	(IV)	Enables glucose transport in cells

Choose the **correct** answer from the options given below:

- (1) A-IV, B-II, C-III, D-I
- (2) A-IV, B-III, C-II, D-I
- (3) A-I, B-IV, C-III, D-II
- (4) A-IV, B-III, C-I, D-II

173. The term optimum pH and optimum temperature for an enzyme-catalyzed reaction refers to the conditions where:

- (1) The enzyme begins to denature.
- (2) The reaction rate is at its minimum.
- (3) The enzyme is most stable.
- (4) The enzyme shows its maximum activity.

174. Consider the structures given below and select a basic amino acid.

- (1) $\begin{array}{c} \text{COOH} \\ | \\ \text{H}-\text{C}-\text{NH}_2 \\ | \\ \text{H} \end{array}$
- (2) $\begin{array}{c} \text{COOH} \\ | \\ \text{H}-\text{C}-\text{NH}_2 \\ | \\ \text{CH}_2-\text{COOH} \end{array}$
- (3) $\begin{array}{c} \text{COOH} \\ | \\ \text{H}_2\text{N}-\text{C}-\text{H} \\ | \\ (\text{CH}_2)_3 \\ | \\ \text{H}_2\text{C}-\text{NH}_2 \end{array}$
- (4) $\begin{array}{c} \text{COOH} \\ | \\ \text{H}-\text{C}-\text{NH}_2 \\ | \\ \text{CH}_2-\text{OH} \end{array}$

175. Read the following statements regarding biomolecules and find the **incorrect** one.

- (1) They are primarily organic, meaning they are built around a carbon framework.
- (2) They encompass a wide range of sizes, from small monomers like glucose to large polymers like starch.
- (3) The term "biomolecule" is restricted to only mean DNA, RNA, and proteins.
- (4) They are essential for the metabolic, structural, and information-storage functions of a living cell.

176. Given below are two statements.

Statement I: The right end of a polysaccharide chain is reducing and left end of insulin is N-terminal.

Statement II: The starch in plants acts as the store house of energy.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

177. Consider the following statements regarding the polysaccharide chitin:

- A. It is a homopolymer.
- B. Its primary biological function is to serve as a readily available source of energy for cells.
- C. It is a key structural component found in the cell walls of fungi and the exoskeletons of arthropods.
- D. It is a soft, fibrous protein responsible for the flexibility of cell membranes.

How many of the above statements are **correct**?

- (1) One
- (2) Two
- (3) Three
- (4) Four

178. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Niacin	(I)	Represented by acid soluble pool
(B)	Zinc	(II)	Present in coenzyme NAD
(C)	Glucose	(III)	Co-factor for carboxypeptidase
(D)	Cytoplasm of cell	(IV)	Have same number of carbons as in fructose

Choose the **correct** answer from the options given below:

- (1) A-II, B-I, C-III D-IV
- (2) A-II, B-III, C-IV, D-I
- (3) A-I, B-II, C-III, D-IV
- (4) A-II, B-IV, C-I, D-III

179. A certain chemical reaction proceeds at a rate of $2.4 \text{ mol L}^{-1}\text{s}^{-1}$ at 30°C .

Assuming the general rule of thumb for the reaction rate w.r.t temperature, what will be the rate of the reaction at 60°C ?

- (1) $4.8 \text{ mol L}^{-1}\text{s}^{-1}$
- (2) $9.6 \text{ mol L}^{-1}\text{s}^{-1}$
- (3) $19.2 \text{ mol L}^{-1}\text{s}^{-1}$
- (4) $38.4 \text{ mol L}^{-1}\text{s}^{-1}$

180. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: When the substrate concentration is equal to the K_m , the rate of the enzymatic reaction is $V_{\max} / 2$.

Reason R: The Michaelis-Menten constant (K_m) is a measure of the substrate concentration at which the enzyme is fully saturated.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Both A and R are true and R is the correct explanation of A.
- (2) Both A and R are true but R is not the correct explanation of A.
- (3) A is true but R is false.
- (4) A is false but R is true.



SPACE FOR ROUGH WORK



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Test Analysis Sheet

- Student Name :
- PW Registration No. :
- Batch Code :
- Test Name/Type :

Max. Marks:
Marks Obtained:
Percentage:
Date of Exam :

1. Subject-wise Performance

Subject	Attempt Order	Time taken (min)	Marks Expected	No. of Attempted Q's	No. of Wrong Q's	Accuracy (%)
Physics						
Chemistry						
Botany						
Zoology						

- Was there a rush at the end? (Yes/No): _____
- Did you revise after solving? (Yes/No): _____

2. Wrong Questions Analysis

Reason	Physics	Chemistry	Botany	Zoology	Remark
Calculation mistake					
Silly mistake					
Marked incorrect option, however solved correctly					
Conceptual Error					
Others					

3. Unattempted Questions Analysis

Reason	Physics	Chemistry	Botany	Zoology	Remark
Forgot the concept / Formula					
Solved correctly but didn't have confidence to mark it on OMR					
Missed the class /Didn't revise					
Got confused with the concept					
Others					

4. Weaknesses Identified

(List topics/chapters where you lost marks)

- Physics :
- Chemistry :
- Botany :
- Zoology :

5. Action Plan Before Next Test

1. Types of Mistakes to Avoid:
2. Extra Practice Needed in:
3. Strategy Changes: (e.g., attempt Chemistry first, manage time better):
4. What would you like to improve emotionally/mentally for the next test? :

Student Signature: _____